Representation of Data

ClipSAT Printable Practice Suite (clipsat.org) • Topic 15.01

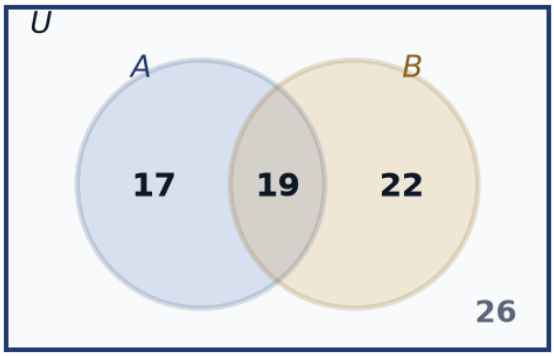
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Stem-and-leaf diagrams

The data set $23, 25, 31, 34, 34, 38, 42, 45$ is to be shown as a stem-and-leaf diagram with stems $2, 3, 4$. Write out the diagram, and use it to find the median.

2 [Foundational] • Target: Stem-and-leaf diagrams

The data set $17, 19, 22, 26, 26, 29, 33, 37, 41$ is to be shown as a stem-and-leaf diagram with stems $1, 2, 3, 4$. Write out the diagram, and use it to find the range.



3 [Foundational] • Target: Histograms and frequency density

A class interval of width 10 contains 25 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

4 [Foundational] • Target: Histograms and frequency density

A class interval of width 4 contains 18 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

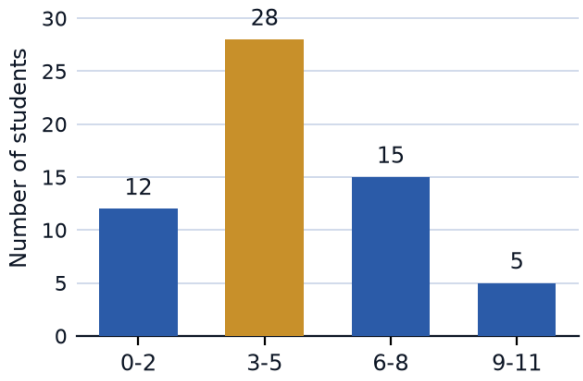
5 [Foundational] • Target: Histograms and frequency density

A class interval of width 5 contains 30 items. Find the frequency density, which is plotted on the vertical axis of a histogram.

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6 [Foundational] • Target: Reading grouped frequency data from a bar chart

The bar chart shows the number of books read by students in a reading challenge, grouped by number of books. Find the total number of students, and the modal group.



7 [Foundational] • Target: Reading grouped frequency data from a bar chart

Using the same reading-challenge data (0-2: 12, 3-5: 28, 6-8: 15, 9-11: 5 students), find what percent of students read 6 or more books.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped data set has class midpoints and frequencies: 5 (freq 4), 15 (freq 8), 25 (freq 6), 35 (freq 2). Estimate the mean.

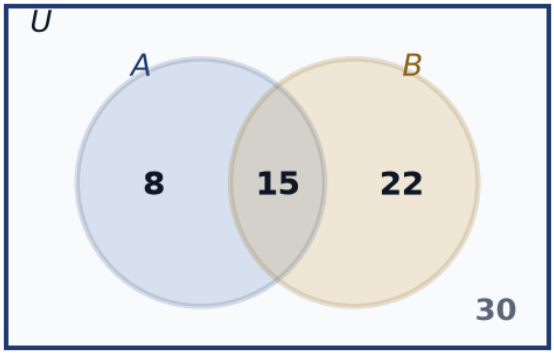
9 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped data set has class midpoints and frequencies: 10 (freq 3), 20 (freq 9), 30 (freq 5), 40 (freq 3). Estimate the mean.

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10 [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 8$, $Q_1 = 15$, median $= 22$, $Q_3 = 30$, and maximum $= 40$. State the interquartile range, and determine whether a value of 45 would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.



11 [Intermediate] • Target: Box-and-whisker plots

A data set has minimum $= 5$, $Q_1 = 12$, median $= 18$, $Q_3 = 27$, and maximum $= 50$. State the interquartile range, and determine whether the maximum value would be considered an outlier if outliers are defined as values beyond $Q_3 + 1.5 \times IQR$.

12 [Intermediate] • Target: Box-and-whisker plots

A data set has $Q_1 = 20$ and $Q_3 = 32$. Find the lower outlier boundary, defined as any value below $Q_1 - 1.5 \times IQR$.

13 [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 42$, median $= 38$, and mode $= 35$. Describe the skewness of the distribution.

14 [Intermediate] • Target: Skewness from summary statistics

A data set has mean $= 28$, median $= 32$, and mode $= 35$. Describe the skewness of the distribution.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

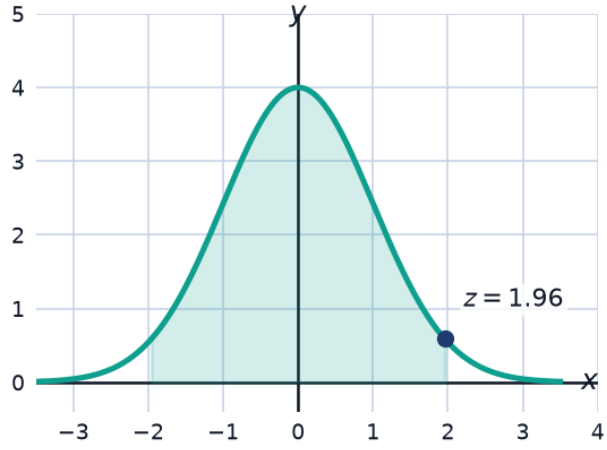
15 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 11.01 Kinematics

Two classes take the same test. Class A has mean 68 and standard deviation 5. Class B has mean 71 and standard deviation 12. Compare the two classes' performance and consistency.

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16 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 12.01 Forces & Equilibrium

Two machines fill bottles with a target volume of 500 mL. Machine X has mean fill volume 500 mL and standard deviation 2 mL. Machine Y has mean fill volume 498 mL and standard deviation 8 mL. Compare the two machines' accuracy and consistency.



17 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 13.01 Newton's Laws

This question has two parts. A delivery company records the delivery times (in minutes) for 50 packages: mean $= 34$ minutes, standard deviation $= 6$ minutes. Company policy states any delivery beyond 2 standard deviations above the mean is classified as late. (a) Find the delivery-time threshold, in minutes, above which a delivery is classified as late. (b) If a customer's package took 44 minutes to deliver, determine, using the standard deviation, whether this delivery should be classified as late, and state how many standard deviations from the mean this value lies.

18 [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 14.01 Energy, Work & Power

A grouped frequency table has class intervals of width 10 starting at 9, with frequencies $[6, 6, 14, 11, 12]$ respectively. Using midpoints, find the mean.

19 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 16.01 Permutations & Combinations

A grouped frequency table has class intervals of width 8 starting at 11, with frequencies $[5, 10, 9, 11, 14]$ respectively. Using midpoints, find the mean.

20 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 17.01 Discrete Random Variables

A grouped frequency table has class intervals of width 10 starting at 15, with frequencies $[13, 12, 13, 4, 4]$ respectively. Using midpoints, find the mean.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

In an ordered stem-and-leaf diagram of 15 test scores, the key states $3 \mid 5 = 35$. The leaves for stem 4 are $1, 3, 5, 8, 8$. What is the median score if there are 4 scores below stem 4?

a. 45

b. 43

c. 48

d. 41

22 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

A data set has minimum $= 12$, $Q_1 = 20$, median $= 28$, $Q_3 = 36$, and maximum $= 52$. What is the interquartile range (IQR)?

a. 16

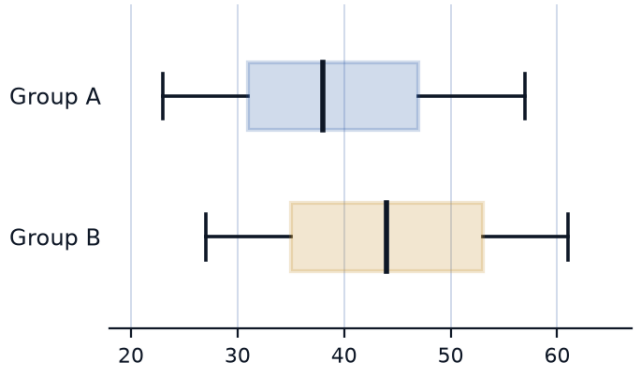
b. 24

c. 40

d. 8

23 [Foundational] • Target: Stem-and-Leaf Diagrams & Box Plots

Using the $1.5 \times$ IQR rule on the data from the previous question, determine if the maximum value 52 is an outlier.



a. No, the upper outlier boundary is 60

b. Yes, any value above 48 is an outlier

c. Yes, any value above 50 is an outlier

d. No, because the range is less than 50

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24 [Foundational] • Target: Histograms & Frequency Density

In a histogram, the class interval $20 \leq x < 35$ has a frequency of 45. What is the frequency density for this bar?

a. 3.0

b. 1.5

c. 4.5

d. 2.25

25 [Foundational] • Target: Histograms & Frequency Density

On a histogram, a bar representing the interval $10 \leq x < 20$ has a width of 2 cm and a height of 5 cm, representing 30 items. What area represents 1 item?

a. $\frac{1}{3}$ cm^2

b. $\frac{1}{2}$ cm^2

c. 3 cm^2

d. 1 cm^2

26 [Foundational] • Target: Histograms & Frequency Density

Why is frequency density, rather than raw frequency, plotted on the vertical axis of a histogram for grouped continuous data?

a. Because class widths may be unequal and area must be proportional to frequency

b. Because continuous data cannot be plotted with integer frequencies

c. To ensure the total height of all bars equals 100

d. To eliminate the need for class boundaries

27 [Foundational] • Target: Cumulative Frequency Curves & Percentiles

When plotting a cumulative frequency graph for grouped continuous data, at which points should the cumulative frequencies be plotted?

a. At the upper class boundaries

b. At the midpoints of each class

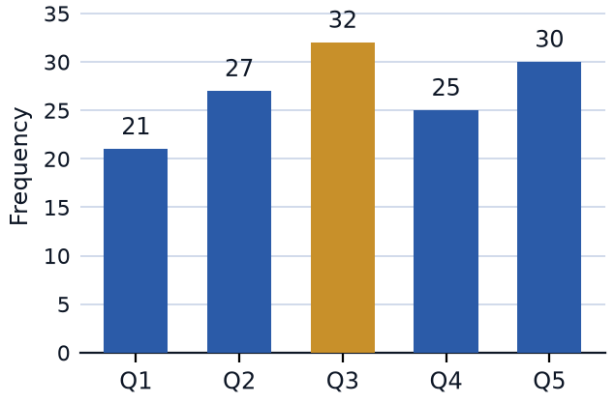
c. At the lower class boundaries

d. At the class frequencies

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28 [Foundational] • Target: Cumulative Frequency Curves & Percentiles

From a cumulative frequency curve of 200 students, the 80th percentile corresponds to which cumulative frequency?



a. 160

b. 80

c. 120

d. 180

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Cumulative Frequency Curves & Percentiles

Using linear interpolation on grouped data, the median lies in the class $40 \le x < 50$ with frequency 25 and cumulative frequency before this class equal to 35. If total $n = 100$, find the estimated median.

a. 46.0

b. 45.0

c. 44.5

d. 47.2

30 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

A data set x has mean $\bar{x} = 50$ and variance $s_x^2 = 16$. A new variable is defined as $y = 3x - 10$. Find the mean $\bar{y}$ and standard deviation s_y.

a. $\bar{y} = 140, s_y = 12$

b. $\bar{y} = 140, s_y = 36$

c. $\bar{y} = 150, s_y = 12$

d. $\bar{y} = 140, s_y = 4$

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31 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

A variable t is coded as $u = \frac{t-200}{10}$. For 50 observations, $\sum u = 85$ and $\sum u^2 = 320$. Find the actual mean of t.

a. 217

b. 215

c. 208.5

d. 220

32 [Intermediate] • Target: Linear Coding & Transformations of Summary Stats

For the data in the previous question, calculate the variance of the original variable t.

a. 351

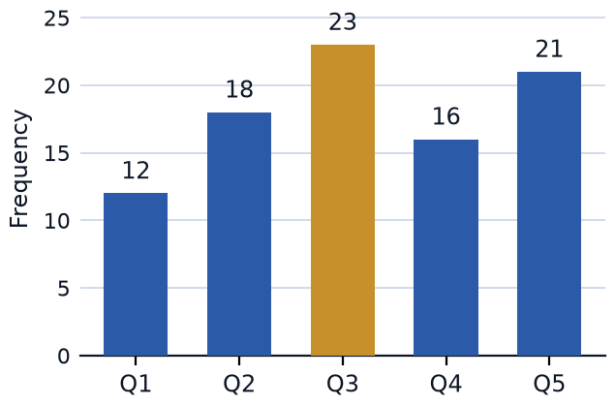
b. 35.1

c. 320

d. 500

33 [Intermediate] • Target: Stem-and-leaf diagrams

A grouped frequency table has class intervals of width 10 starting at 17, with frequencies $[5, 13, 9, 12, 3]$ respectively. Using midpoints, find the mean.



a. 899/21

b. 836/21

c. 857/21

d. 878/21

34 [Intermediate] • Target: Histograms and frequency density

A grouped frequency table has class intervals of width 8 starting at 9, with frequencies $[15, 9, 11, 4, 5]$ respectively. Using midpoints, find the mean.

a. 269/11

b. 280/11

c. 258/11

d. 291/11

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35 [Intermediate] • Target: Reading grouped frequency data from a bar chart

A grouped frequency table has class intervals of width 8 starting at 19, with frequencies $[9, 9, 13, 13, 13]$ respectively. Using midpoints, find the mean.

a. 792/19

b. 754/19

c. 811/19

d. 773/19

36 [Intermediate] • Target: Measures of central tendency for grouped data

A grouped frequency table has class intervals of width 8 starting at 17, with frequencies $[3, 15, 12, 8, 12]$ respectively. Using midpoints, find the mean.

a. 944/25

b. 1019/25

c. 994/25

d. 969/25

37 [Intermediate] • Target: Box-and-whisker plots

A grouped frequency table has class intervals of width 8 starting at 11, with frequencies $[14, 9, 12, 7, 10]$ respectively. Using midpoints, find the mean.

a. 370/13

b. 396/13

c. 383/13

d. 409/13

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Skewness from summary statistics • Cross-Lesson: 14.01 Energy, Work & Power

A grouped frequency table has class intervals of width 8 starting at 9, with frequencies $[11, 7, 7, 15, 10]$ respectively. Using midpoints, find the mean.

a. 799/25

b. 774/25

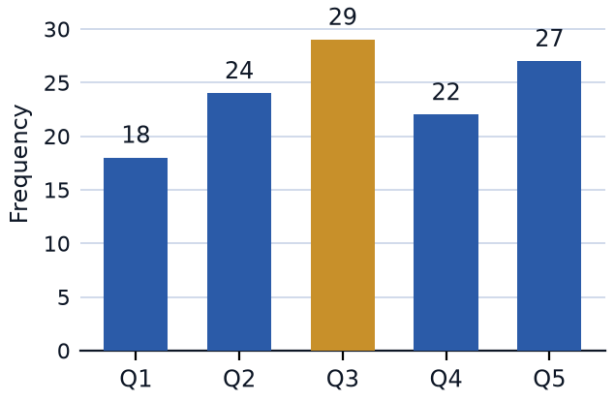
c. 749/25

d. 724/25

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39 [Challenging] • Target: Comparing distributions using summary measures • Cross-Lesson: 16.01 Permutations & Combinations

A grouped frequency table has class intervals of width 8 starting at 6, with frequencies $[6, 6, 11, 7, 10]$ respectively. Using midpoints, find the mean.



a. 144/5

b. 134/5

c. 139/5

d. 149/5

40 [Challenging] • Target: A well-designed word problem: analyzing delivery times • Cross-Lesson: 17.01 Discrete Random Variables

A grouped frequency table has class intervals of width 8 starting at 16, with frequencies $[15, 3, 5, 15, 2]$ respectively. Using midpoints, find the mean.

a. 161/5

b. 171/5

c. 176/5

d. 166/5

41 [Challenging] • Target: Stem-and-Leaf Diagrams & Box Plots • Cross-Lesson: 18.01 Binomial Distribution

A grouped frequency table has class intervals of width 10 starting at 1, with frequencies $[6, 13, 5, 13, 13]$ respectively. Using midpoints, find the mean.

a. 144/5

b. 154/5

c. 149/5

d. 139/5

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42 [Challenging] • Target: Histograms & Frequency Density • Cross-Lesson: 19.01 Normal Distribution

A grouped frequency table has class intervals of width 10 starting at 8, with frequencies $[9, 7, 9, 2, 3]$ respectively. Using midpoints, find the mean.

a. 79/3

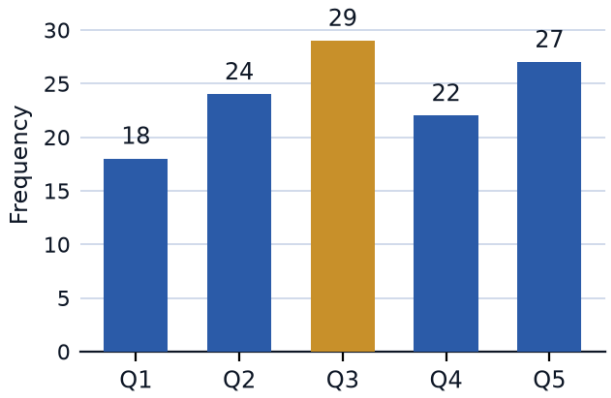
b. 88/3

c. 82/3

d. 85/3

43 [Challenging] • Target: Cumulative Frequency Curves & Percentiles • Cross-Lesson: 2.01 Simultaneous Equations & Inequalities

A grouped frequency table has class intervals of width 10 starting at 9, with frequencies $[6, 7, 13, 7]$ respectively. Using midpoints, find the mean.



a. 323/11

b. 334/11

c. 345/11

d. 356/11

44 [Challenging] • Target: Linear Coding & Transformations of Summary Stats • Cross-Lesson: 3.01 Graphs & Transformations

A grouped frequency table has class intervals of width 10 starting at 10, with frequencies $[15, 6, 6, 6, 7]$ respectively. Using midpoints, find the mean.

a. 32

b. 33

c. 30

d. 31

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45 [Challenging] • Target: Stem-and-leaf diagrams • Cross-Lesson: 4.01 Coordinate Geometry

A grouped frequency table has class intervals of width 10 starting at 11, with frequencies $[14, 13, 4, 3]$ respectively. Using midpoints, find the mean.

a. 405/17

b. 422/17

c. 456/17

d. 439/17

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